

Quiz 3 Section 1 Grading

Context: A manufacturer produces LED bulbs; each bulb has probability $p = 0.03$ of being defective.

Tasks and Full Answers (showing work)

Task 1 — Distribution (5 pts)

Question: Write down the probability distribution for the number X of defectives in n independent trials, where p is the defect probability for one bulb.

Answer (work): The number of defective bulbs follows a **binomial distribution**. Define X = number of defectives in n bulbs. The probability mass function is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Task 2 — Exactly 2 defectives in 15 (5 pts)

Question: Find $P(X = 2)$ for $n = 15$, $p = 0.03$.

Step-by-step work: 1. Use the binomial PMF with $n = 15$, $k = 2$, $p = 0.03$:

$$P(X = 2) = \binom{15}{2} (0.03)^2 (0.97)^{13}.$$

2. Compute components:

- $\binom{15}{2} = 105$
- $(0.03)^2 = 0.0009$
- $0.97^{13} \approx 0.6730270902$

3. Multiply:

$$P(X = 2) = 105 \times 0.0009 \times 0.6730270902 \approx 0.06360106.$$

Final (rounded):

$P(X = 2) \approx 0.06360 \text{ (6.360\%)}$

Task 3 — At most 1 defective in 15 (5 pts)

Question: Find $P(X \leq 1)$ for $n = 15$, $p = 0.03$.

Step-by-step work: We compute $P(X \leq 1) = P(X = 0) + P(X = 1)$.

1. $P(X = 0) = (1 - p)^{15} = 0.97^{15} \approx 0.6332511891$.
2. $P(X = 1) = \binom{15}{1} (0.03)(0.97^{14}) = 15 \times 0.03 \times 0.97^{14} \approx 0.2937763249$.
3. Sum:

$$P(X \leq 1) \approx 0.6332511891 + 0.2937763249 = 0.9270275140.$$

Final (rounded):

$P(X \leq 1) \approx 0.92703 \text{ (92.703\%)}$

Task 4 — Poisson approximation for n=150 (5 pts)

Question: Using the Poisson approximation to the binomial, estimate $P(X = 2)$ when $n = 150$, $p = 0.03$.

Justification: Poisson approximation is appropriate when n is large, p is small, and $\lambda = np$ is moderate.

Step-by-step work: 1. Compute $\lambda = np = 150 \times 0.03 = 4.5$. 2. Poisson formula for $k = 2$:

$$P(X = 2) \approx e^{-\lambda} \frac{\lambda^2}{2!} = e^{-4.5} \frac{4.5^2}{2}.$$

3. Numerics:

- $e^{-4.5} \approx 0.01110899654$
- $\frac{4.5^2}{2} = 10.125$

4. Multiply:

$$P(X = 2) \approx 0.01110899654 \times 10.125 \approx 0.11247859.$$

Final (rounded):

$P(X = 2) \approx 0.11248 \text{ (11.248\%)}$

Grading Rubric (one page — total 20 pts)**General guidelines**

- Each task is worth **5 points**. Total = **20 points**.
- Show work for full credit. Correct numerical answers with no work may receive up to **50%** credit if the value is correct and consistent with the problem.
- Accept rounding differences within ± 0.001 of the provided numeric answers for full credit.

Task 1 (5 pts)

- **2 pts** — Correctly identify the distribution as *binomial* and clearly define the random variable (e.g., X = number of defectives).
- **2 pts** — Correct PMF: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$.
- **1 pt** — State domain $k = 0, 1, \dots, n$ (or equivalent wording).

Common errors and deductions: - Writing the PMF with wrong exponents or swapped p and $1 - p$: -2 pts. - Omitting domain: -1 pt.

Task 2 (5 pts)

- **2 pts** — Correct substitution into PMF with $n = 15$, $k = 2$, $p = 0.03$.
- **1 pt** — Correct computation of $\binom{15}{2} = 105$ (or equivalent demonstration).
- **2 pts** — Correct numeric value (within ± 0.001 of 0.063601).

Partial credit examples: - Correct formula but arithmetic mistakes: award **3 pts**. - Using an approximation method with correct procedure and close value: up to **4 pts**.

Task 3 (5 pts)

- **1 pt** — Recognize $P(X \leq 1) = P(X = 0) + P(X = 1)$ (or equivalent complement approach).
- **2 pts** — Correct formulas and substitutions for $P(X = 0)$ and $P(X = 1)$ with $n = 15$, $p = 0.03$.
- **2 pts** — Correct numeric sum (within ± 0.001 of 0.927028).

Partial credit examples: - If only $P(X = 0)$ is correct: award **1–2 pts**. - If student uses complement $1 - P(X \geq 2)$ and computes correctly: treat as full-credit-equivalent.

Task 4 (5 pts)

- **1 pt** — State Poisson approximation and brief justification (e.g., n large, p small, $\lambda = np$ moderate).
- **1 pt** — Correctly compute $\lambda = np = 4.5$.
- **2 pts** — Correct Poisson formula for $k = 2$ and substitution: $e^{-\lambda} \lambda^k / k!$.
- **1 pt** — Correct numeric evaluation (within ± 0.001 of 0.112479).

Partial credit examples: - If student computes exact binomial for $n = 150, p = 0.03$ but doesn't use Poisson: award **2–3 pts** for correct binomial work and note the response did not follow the approximation request.

Short answer key (for quick grading)

- **T1:** $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.
 - **T2:** $P(X = 2) = \binom{150}{2} (0.03)^2 (0.97)^{148} \approx 0.06360$.
 - **T3:** $P(X \leq 1) = 0.97^{150} + 150(0.03)(0.97^{149}) \approx 0.92703$.
 - **T4:** $\lambda = 4.5, P(X = 2) = \frac{e^{-4.5} 4.5^2}{2!} \approx 0.11248$.
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End of grading document.